

Qualifying Exam

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Short Summary – 8 paper

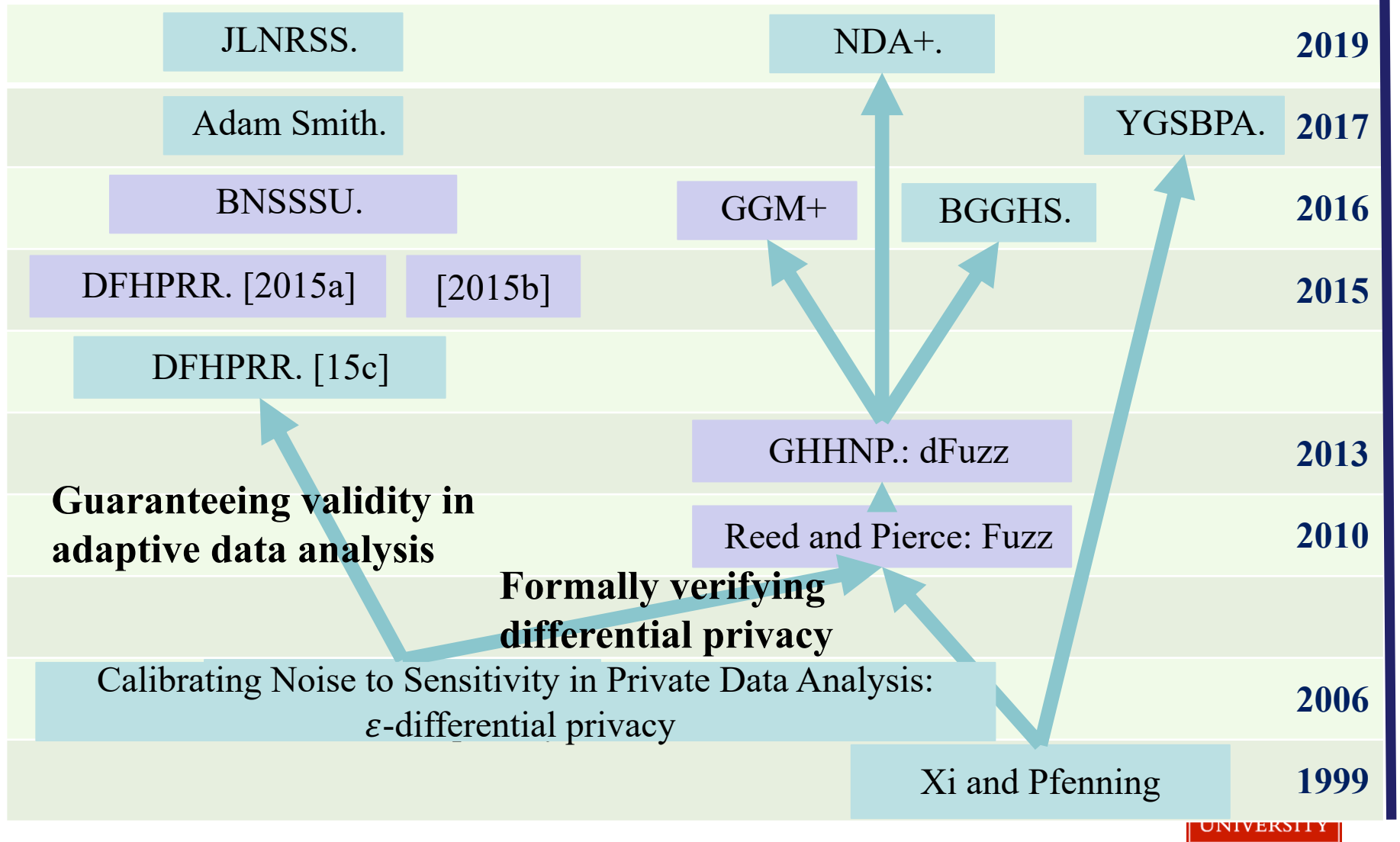
■ Statistical Analysis

- **[DMNS 2006]** – ϵ -differential privacy
- **[DFH 2015c]** – preserving statistical validity in adaptive data analysis through differentially private techniques
- **[Smith 2017]** – survey on adaptive data analysis
- **[JLN 2019]** – new analysis on the differential privacy's generalization

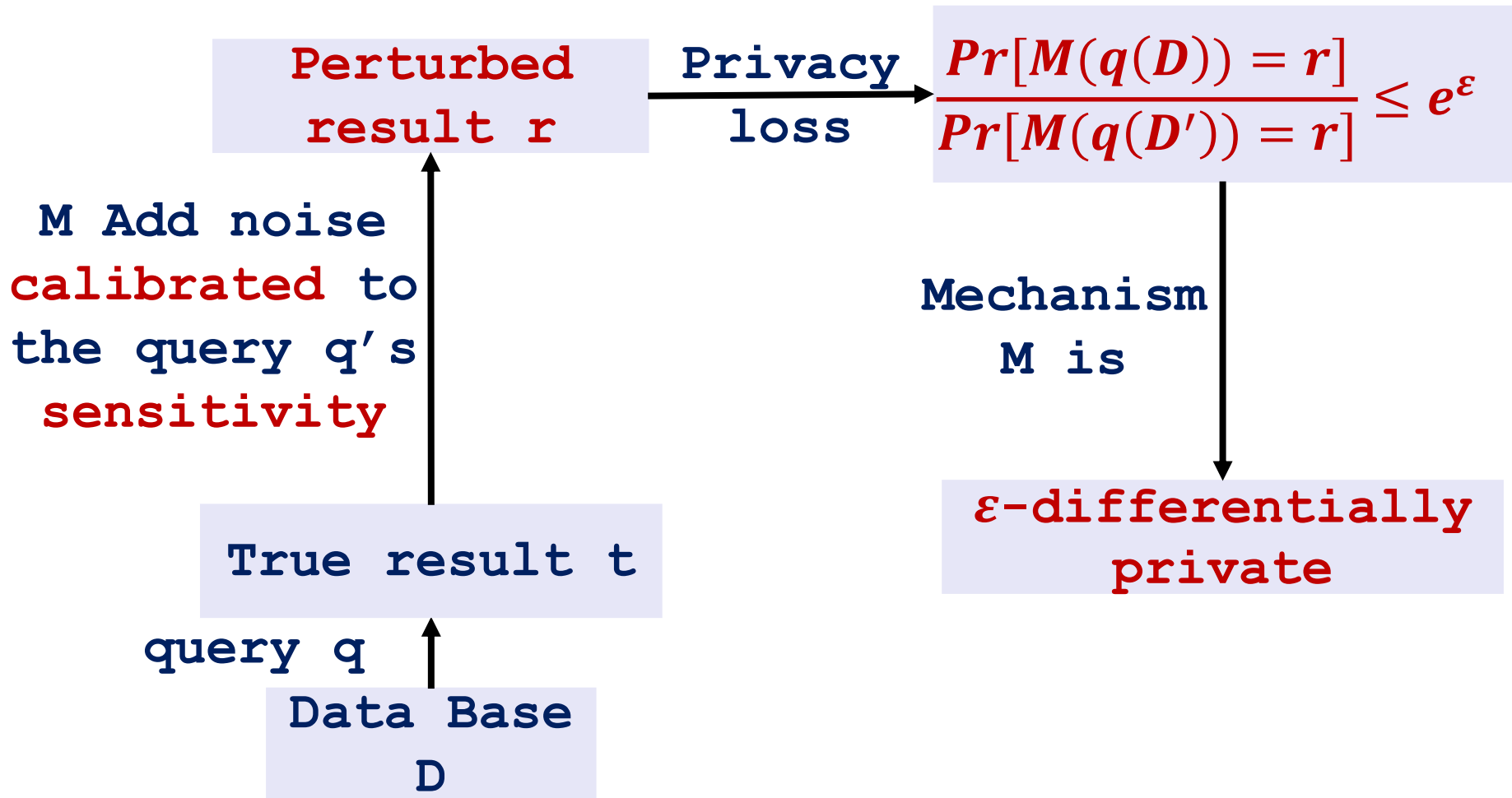
■ Programming Language

- **[XP 1999]** – dependent type system with type index
- **[BGGHS. 2016]** – formally verify differential privacy based on the probabilistic coupling
- **[YGSBPA. 2017]** – verifying the correctness and randomness of HMAC implementation
- **[NDA+. 2019]** – high order language and linear type system for auto verifying the differential privacy

Relations & Importance



Calibrating Noise to Sensitivity in Private Data Analysis:
 ϵ -differential privacy



Guaranteeing validity in adaptive data analysis

JLNRSS.

2019

Adam Smith.

2017

BNSSSU.

2016

DFHPRR. [2015a, b]

2015

Preserving Statistical Validity in Adaptive Data Analysis

2013

First Relevant Direction: Extend dp into
validity Guaranteeing in adaptive data analysis

2010

DMNS.:
 ϵ -differential privacy

2006

1999

Preserving Statistical Validity in Adaptive Data Analysis

- Transfer Theorem – query q is ε –differentially private

Sample
 S

$q(S)$

Distribution
 P

$q(P)$

$\sim$

$$P[q(s) \leq q(p) - \alpha] \leq \beta/2$$

$$\varepsilon \leq \alpha/2, \quad \alpha \geq \sqrt{12 \ln(4/\beta)/n}$$

Guaranteeing validity in adaptive data analysis

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$$P[|q(s) - q(p)| \geq \epsilon] \leq e^{-O(\epsilon^2 n)}$$

Holdout Reuse

Guaranteeing validity in adaptive data analysis

JLNRSS.

2019

Information Privacy and Stability in Adaptive Data Analysis

2017

BNSSSU.

2016

DFHPRR. [2015a, b]

2015

DFHPRR. [15c]

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DMNS.:
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Information Privacy and Stability in Adaptive Data Analysis

- On Generalization Error in **Linear Data analysis**
 - Straightforward approach
 - k nonadaptive queries – $\Theta(\sqrt{\log(k)/n})$
 - k adaptive queries – $\Omega(\sqrt{k/n})$
 - Data Splitting - k adaptive queries – $\Theta(\sqrt{k(\log(k))/n})$
 - Known Bounds
 - efficient mechanism with k queries – $O(\sqrt[4]{k}/\sqrt{n})$
 - r adaptive and k nonadaptive queries – $O(\sqrt{r(\log(k))/n})$
 - inefficient mechanism – $O(\sqrt[6]{\log(N)}\sqrt[3]{(\log(k))/n})$
 - Bounds for differentially private algorithms
 - (ϵ, δ) -differential privacy – $(O(\epsilon), O(\delta))$ - accurate
- Generalization Error in **Non-Linear Data analysis**
 - Based on Information Measures

Guaranteeing validity in adaptive data analysis

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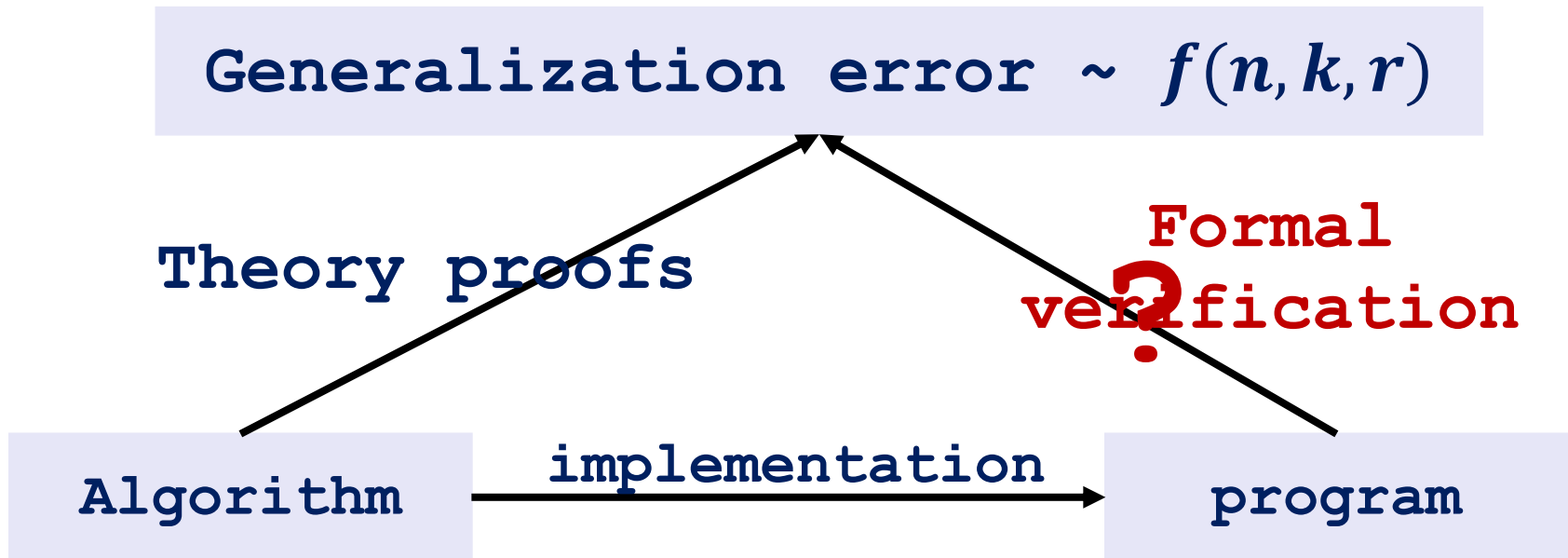
2006

1999

Following the same notation, if M is (ϵ, δ) -differentially private, q is selected by M , c is any constant:

$$P[\max |q(s) - q(p)| \geq (e^\epsilon - 1) + 2c] \leq \frac{\delta}{c}$$

Consideration



Formal Verification of differential privacy

Enriched type system with dependent types:

- Index objects from Constraint Domains:

Index sort: γ

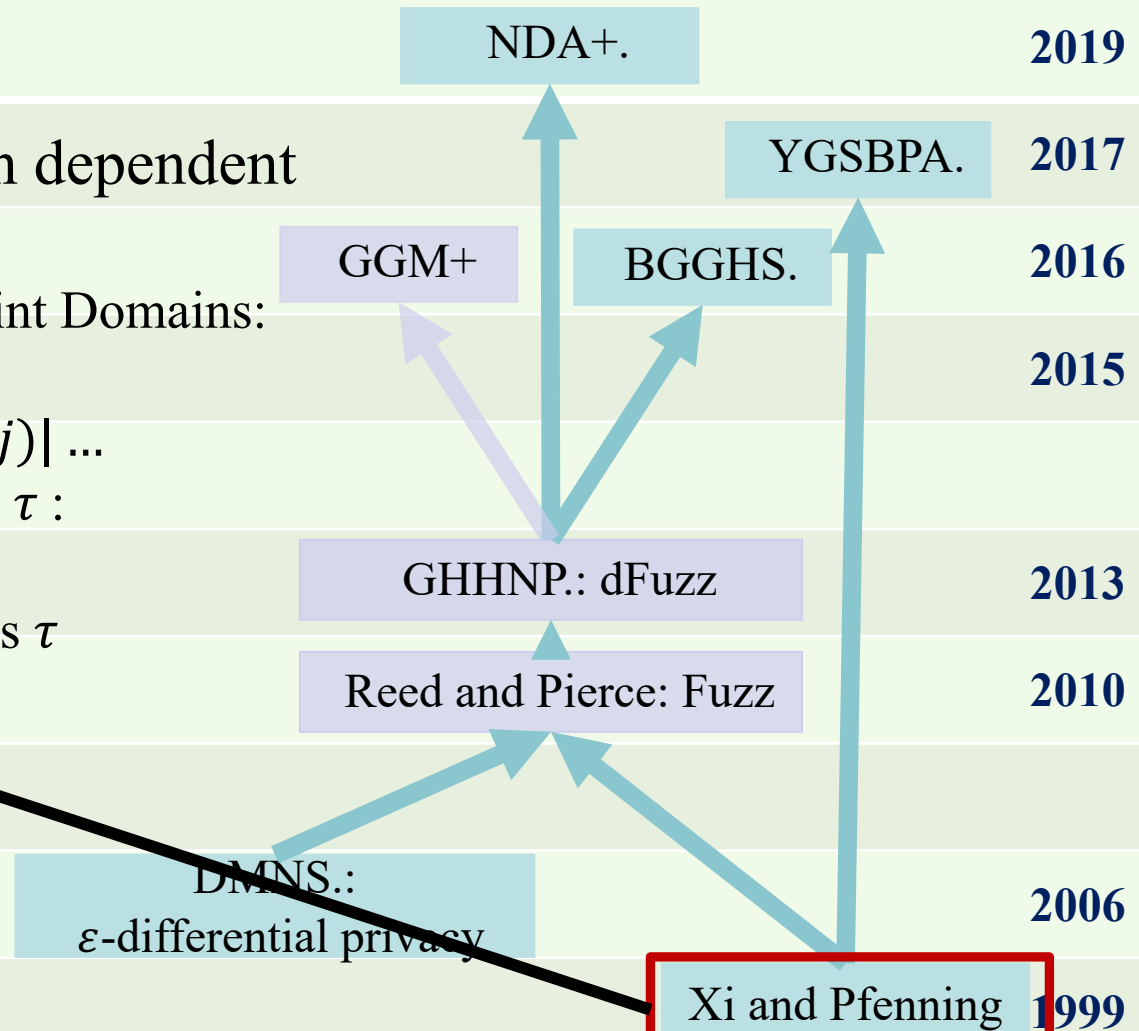
Index objects: $a \mid () \mid (i, j) \mid \dots$

- Universal Dependent Types τ :

$\Pi a: \gamma. \tau$

- Existential Dependent Types τ

$\Sigma a: \gamma. \tau$



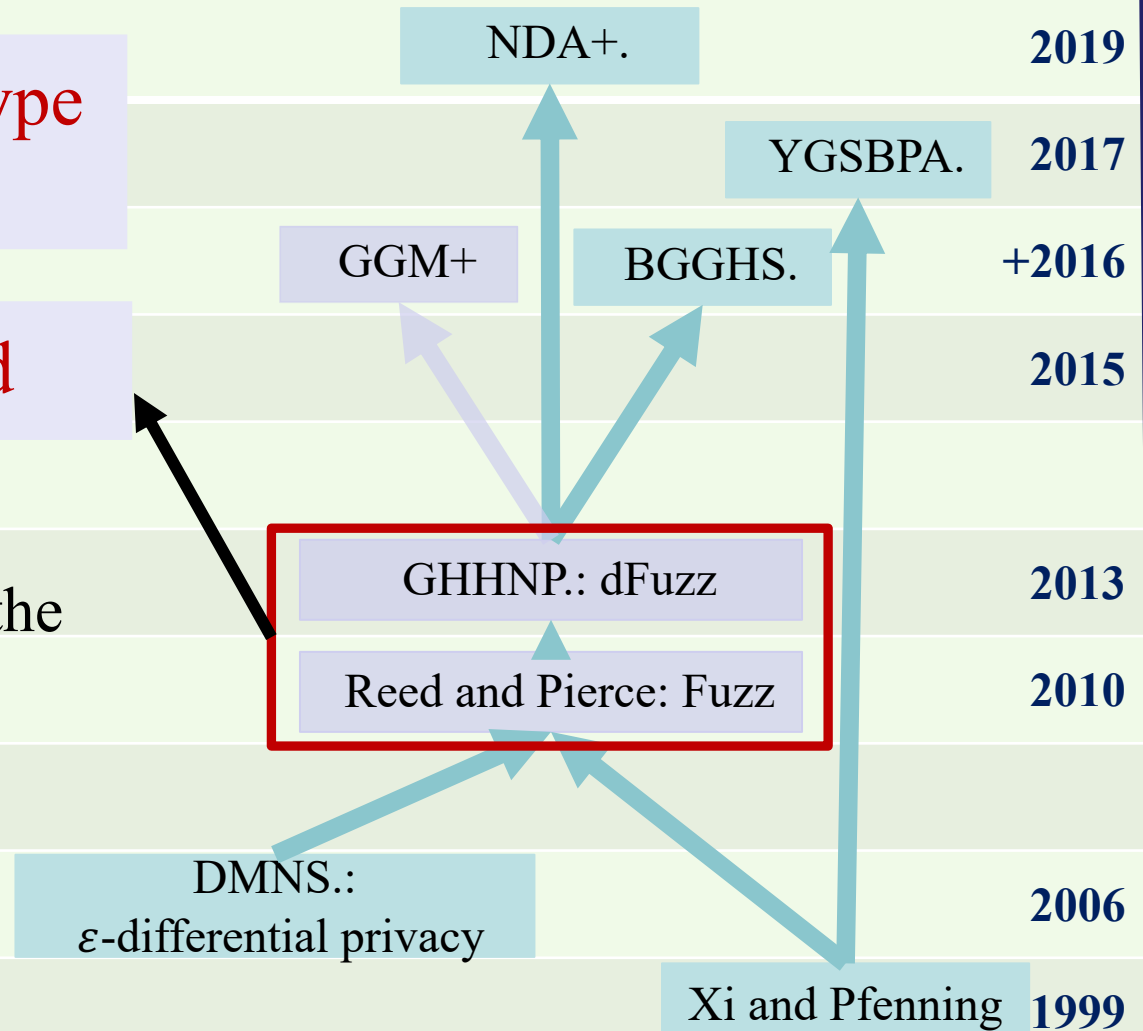
Formal Verification of differential privacy

Linear Dependent Type
System

+

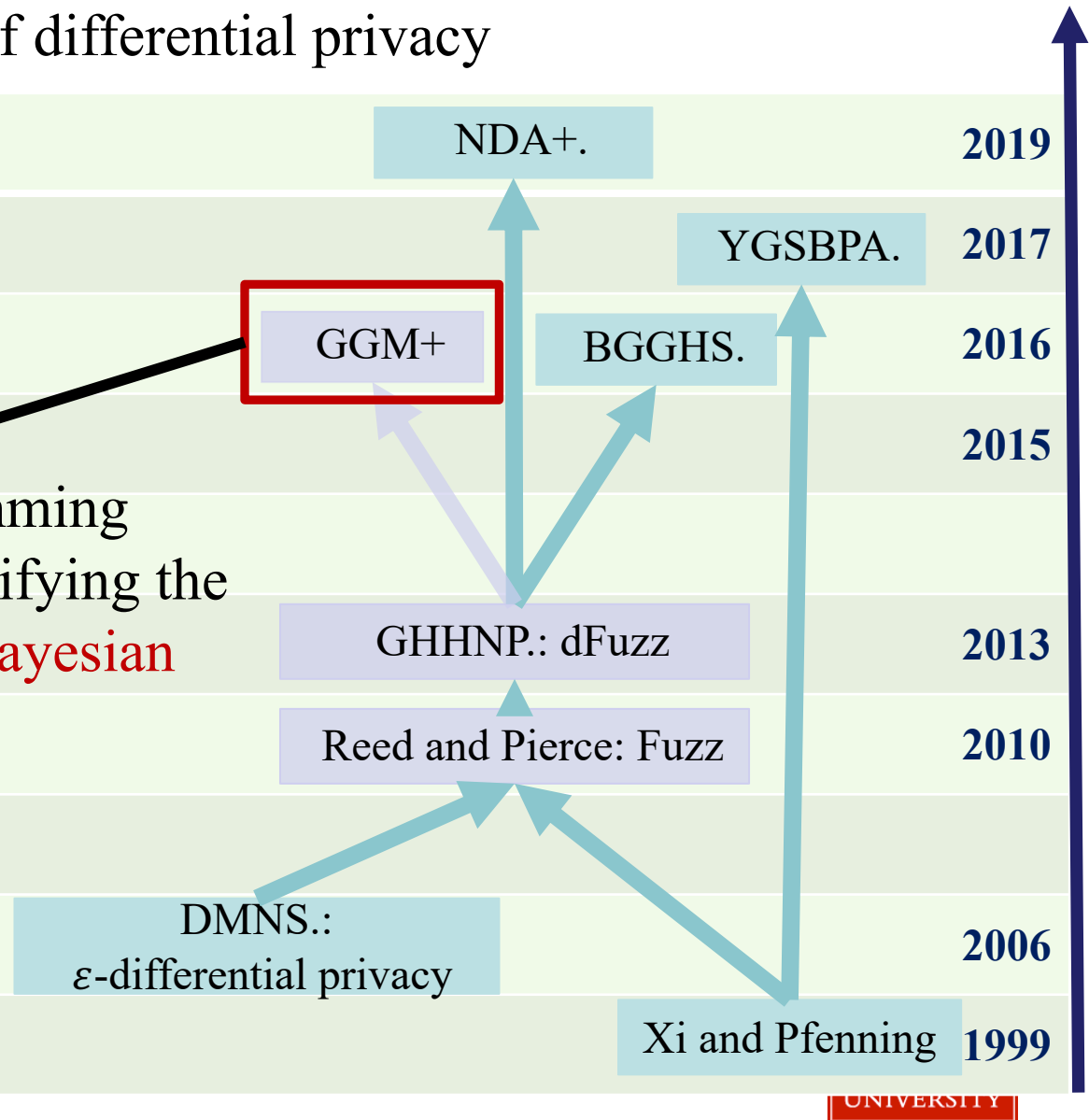
Probability monad

Track the **sensitivity** and
automatically guarantee the
differential privacy



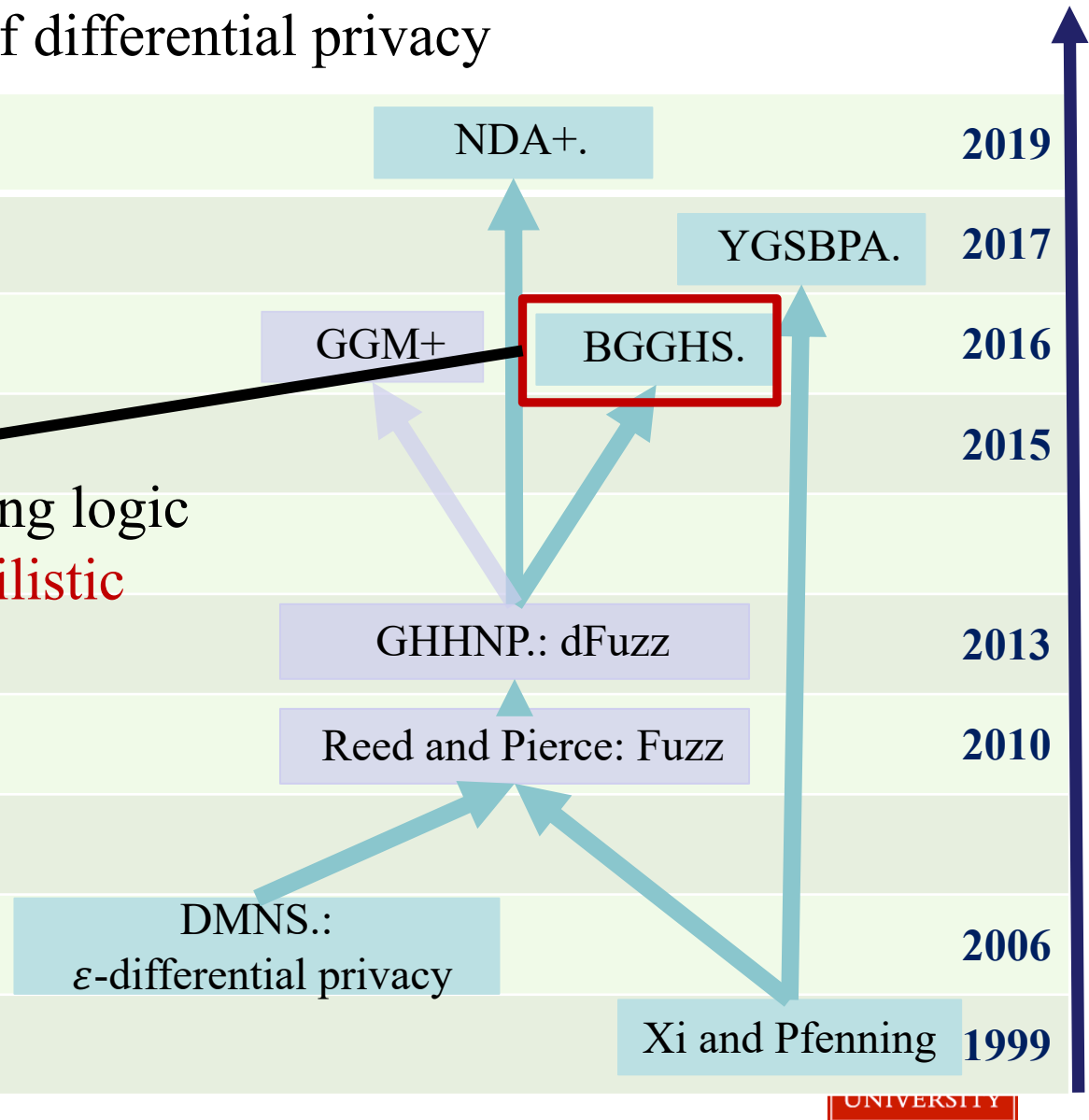
Formal Verification of differential privacy

A probabilistic programming
system for formally verifying the
**differentially private Bayesian
inference**



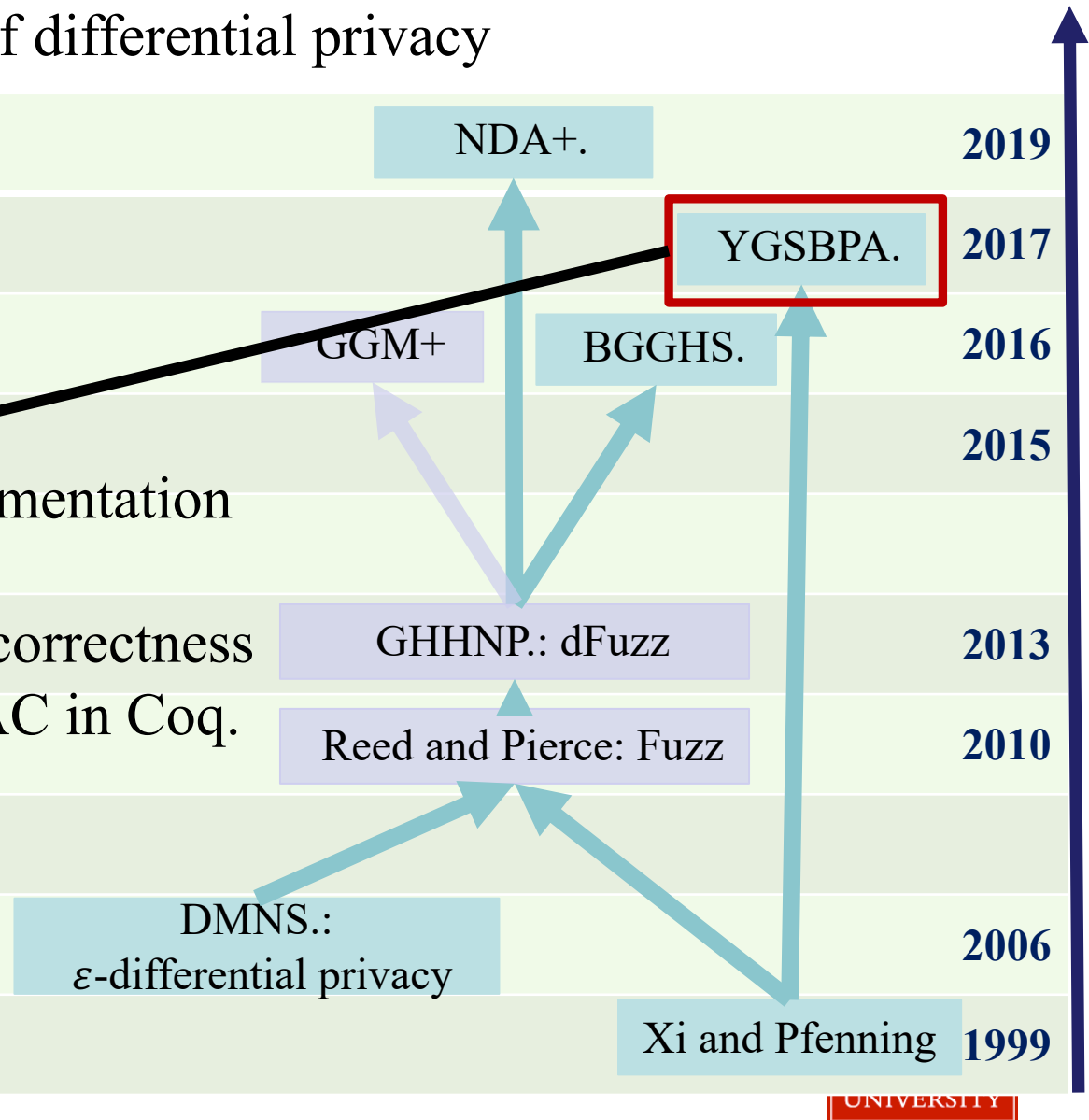
Formal Verification of differential privacy

A relational programming logic
combining with **probabilistic
couplings and liftings**.

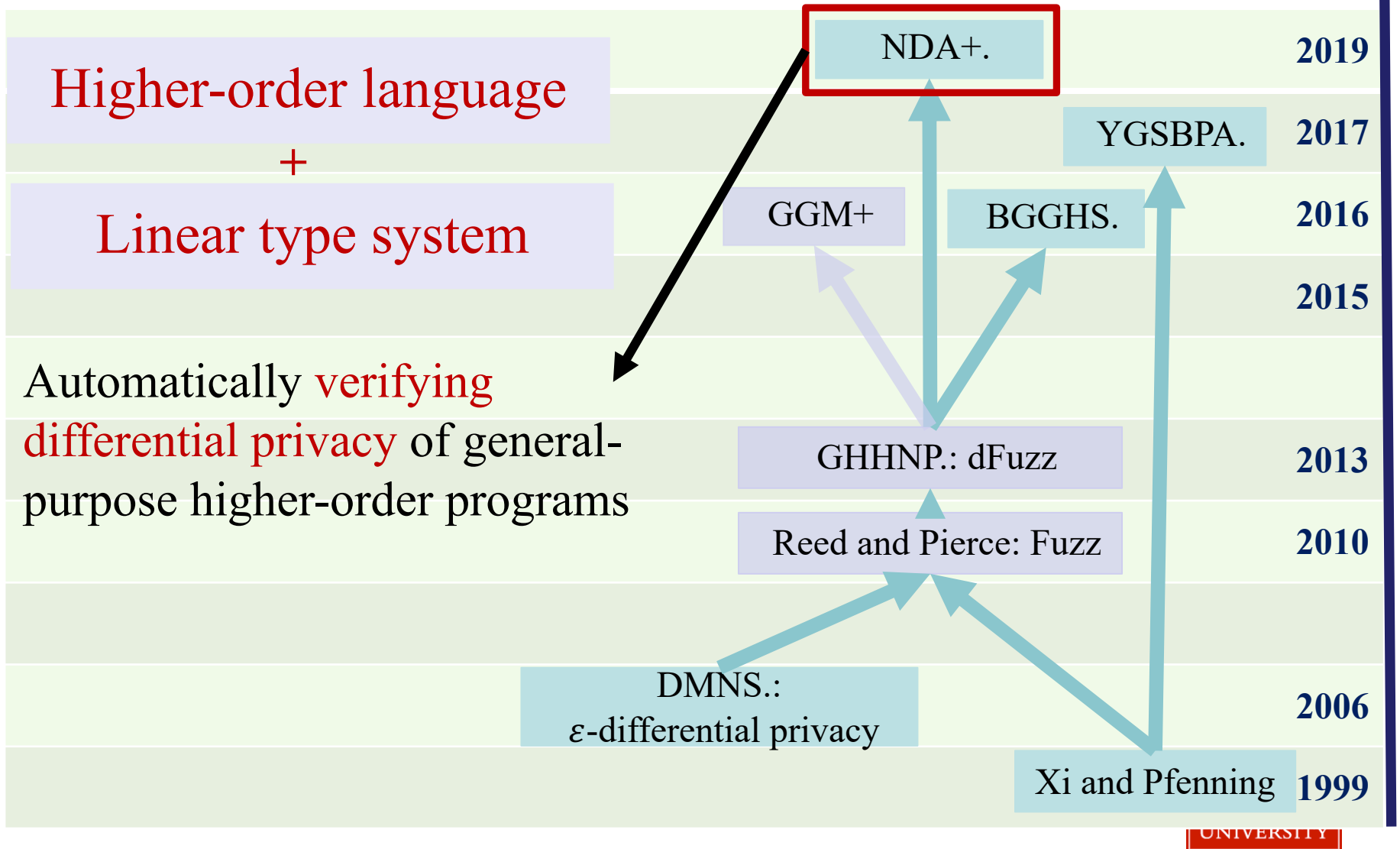


Formal Verification of differential privacy

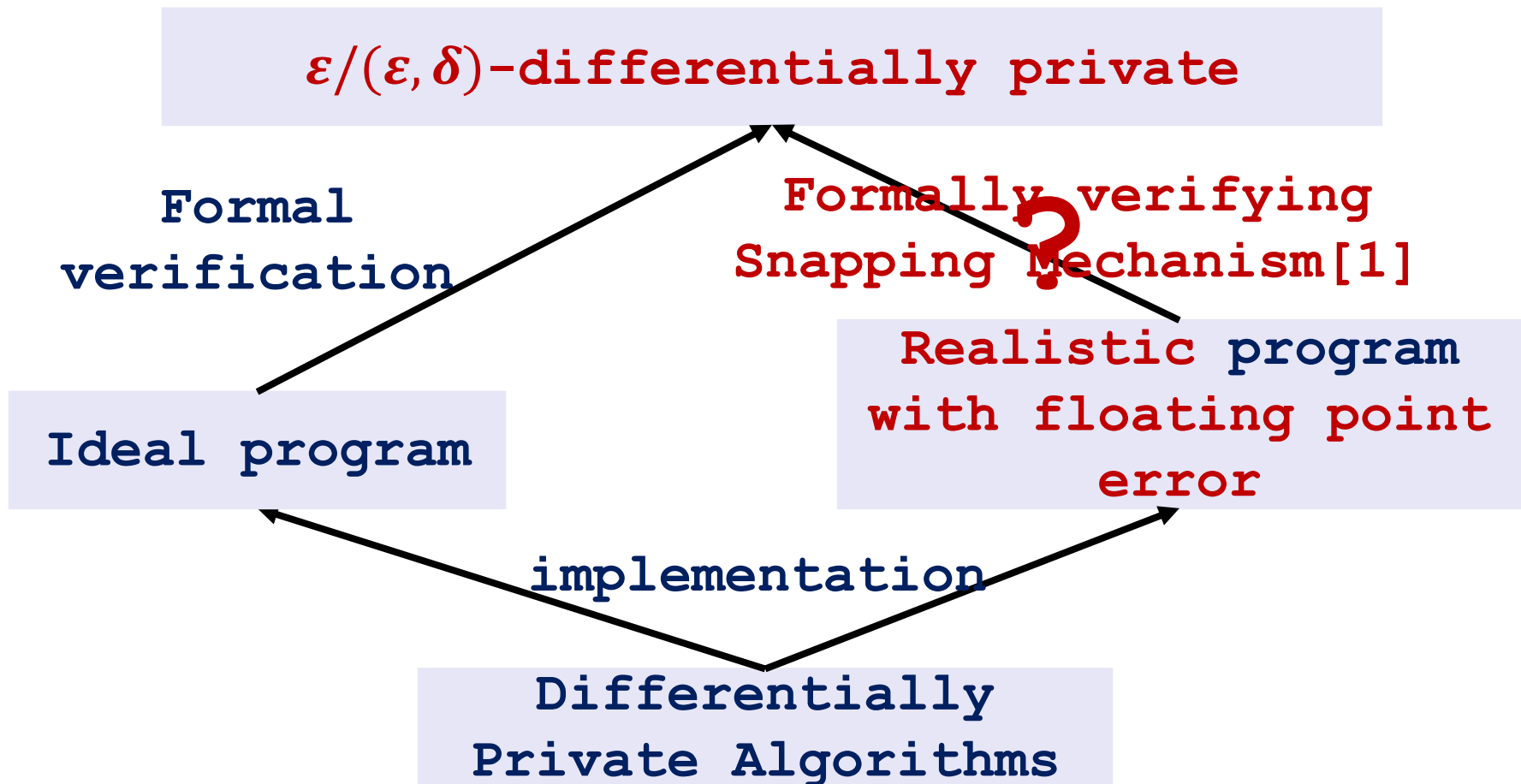
1. Formalizing the implementation of HMAC
2. Formally verified the correctness and randomness of HMAC in Coq.



Formal Verification of differential privacy



Consideration



Possible Research Directions

formal verification of the program's statistical validity.

Formally verifying the realistic implementation of dp-algorithms: snapping mechanism

Guaranteeing validity in adaptive data analysis

JLNRSS.

Adam Smith.

BNSSSU.

DFHPRR. [2015a] [2015b]

DFHPRR. [15c]

DMNS.: ϵ -differential privacy

Formally verifying differential privacy

NDA+.

2019

YGSBPA.

2017

GGM+

BGGHS.

2016

2015

GHHNP.: dFuzz

2013

Reed and Pierce: Fuzz

2010

Xi and Pfenning

2006

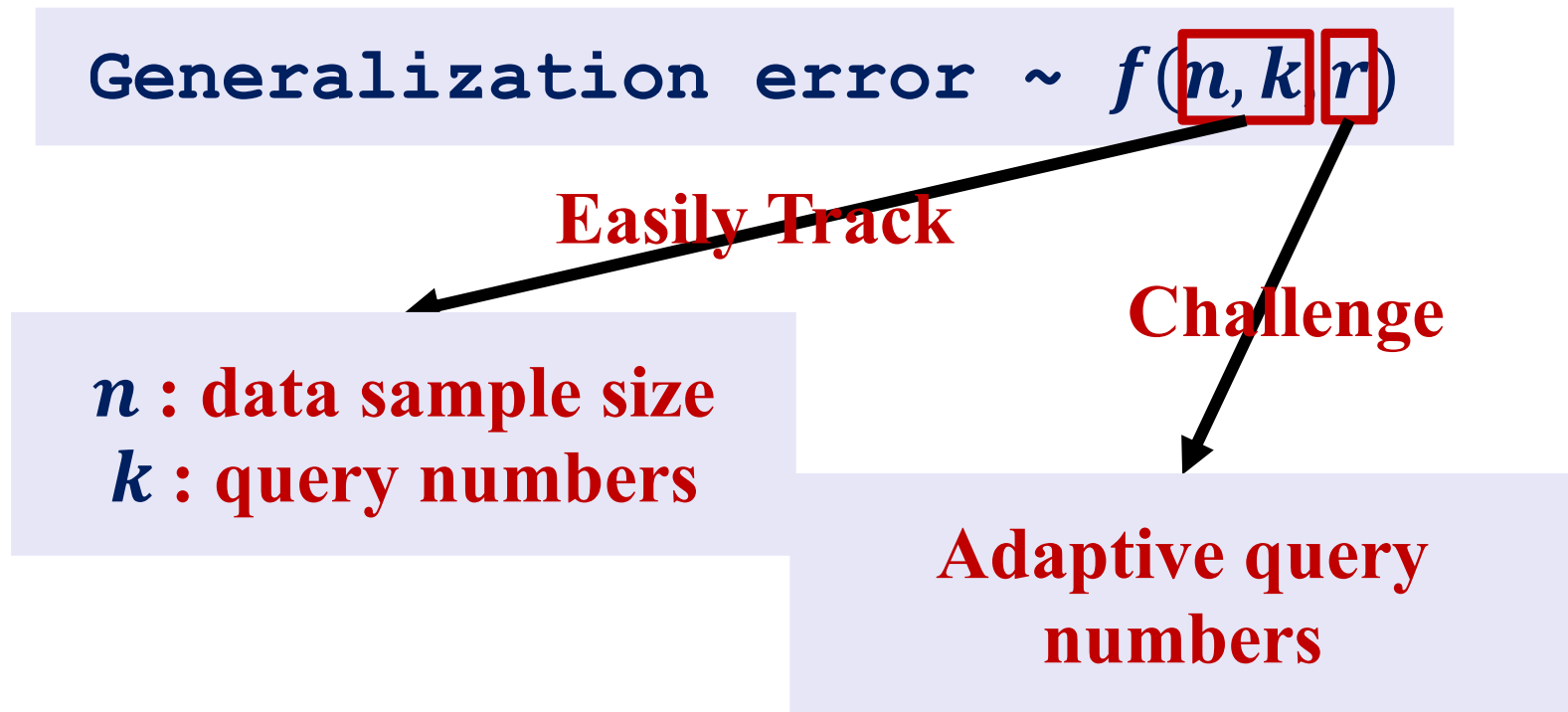
1999

UNIVERSITY

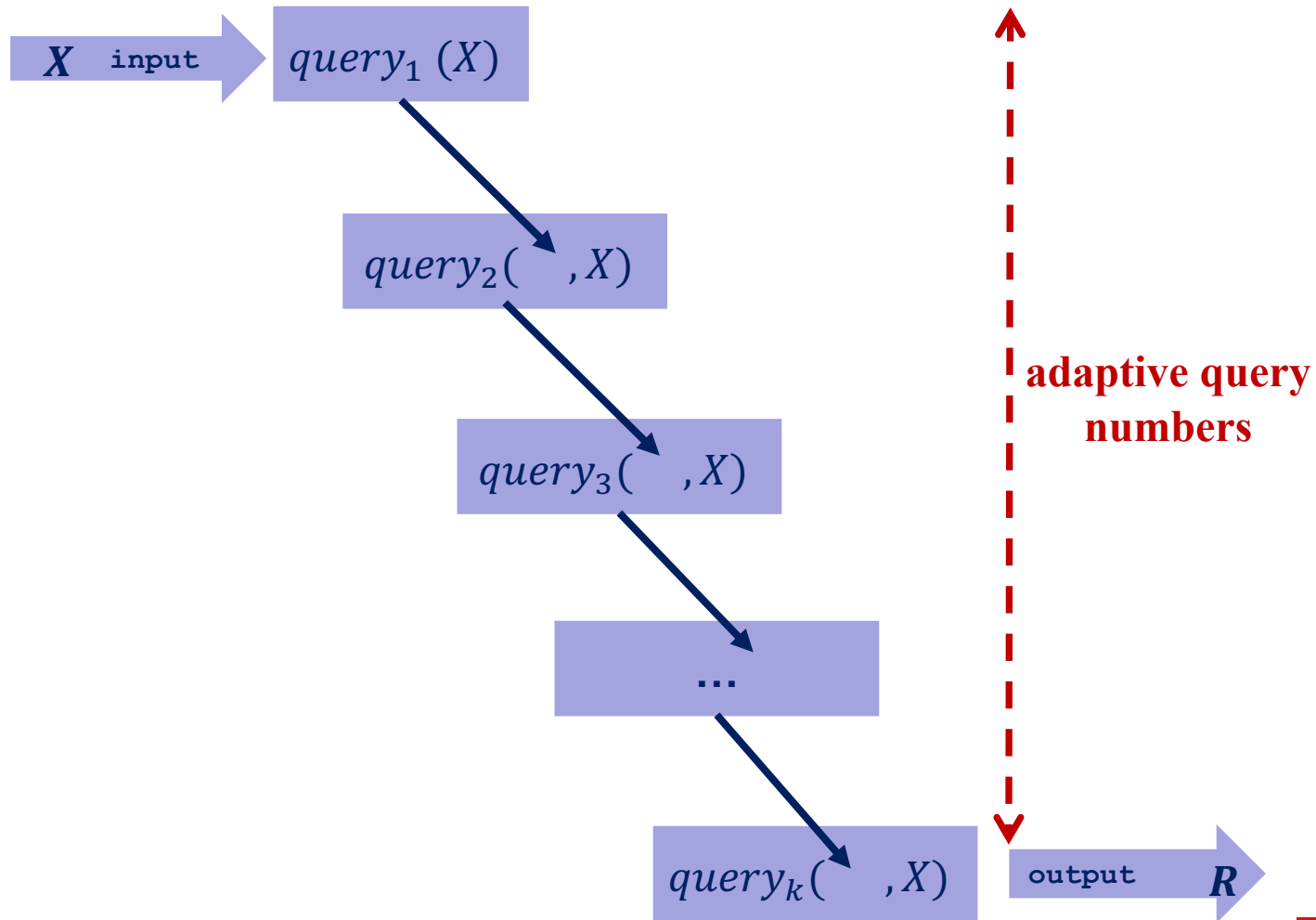
Formal verification of the program's statistical validity.

- **Research Goals.** [DFHPRR.'15, Adam Smith'17, JLNRSS.'19...]
 - Guaranteeing the generalization error for implementations of adaptive data analysis.
 - Helping programmer to optimize their programs to have better generalizations.
- **Challenges.**
 - Figuring out the program's adaptive query numbers, i.e. the program's adaptivity.
- **Methodologies.** [Xi and Pfenning '99 , NDA+'19, ...]
 - Designing a dependent type system and language that

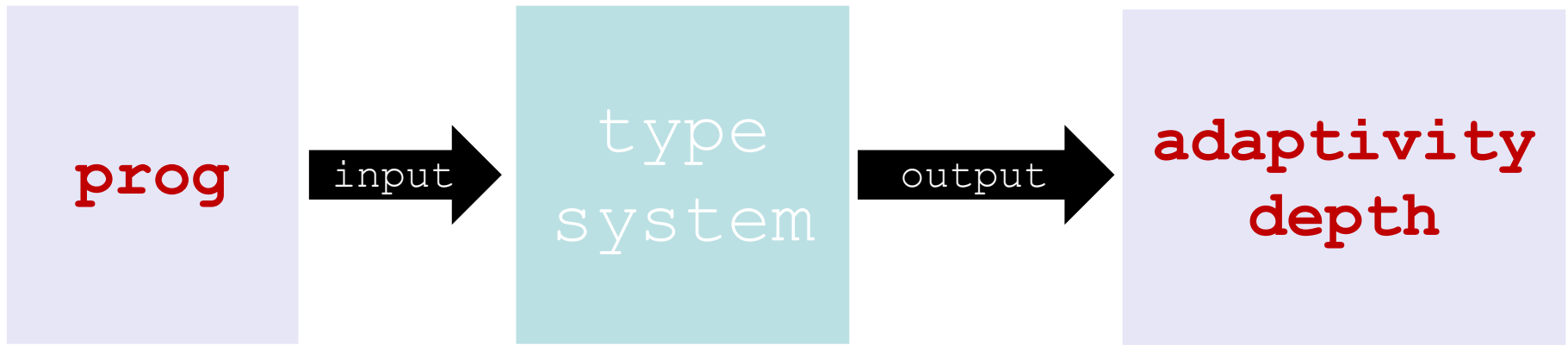
Research Goals &. Challenge



Research Goals &. Challenge



Methodologies



Methodologies – Language Design

- Refinement Types [Xi and Pfenning 1999]
 - Helping programmer to optimize their programs to have better generalizations.
- Expressions. [LG '11...]
 - Figuring out the program's adaptive query numbers, i.e. the program's adaptivity.
- Typing Judgement. [LG '11...]
 - approximates an **upper bound** on adaptive query numbers

$$\vdash^z e : \tau$$

upper bound of adaptive query numbers

Possible Research Directions

formal verification of the program's statistical validity.

Formally verifying the realistic implementation of dp-algorithms: snapping mechanism

Guaranteeing validity in adaptive data analysis

JLNRSS.

Adam Smith.

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DFHPRR. [2015a] [2015b]

DFHPRR. [15c]

Formally verifying differential privacy

NDA+.

2019

YGSBPA.

2017

GGM+

BGGHS.

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2015

GHHNP.: dFuzz

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Reed and Pierce: Fuzz

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DMNS.: ϵ -differential privacy

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Xi and Pfenning

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Formal verification of differential privacy (snapping mechanism)

- **Research Goals.** [Ilya Mironov '12...]
 - Verifying the actual privacy loss for the realistic implementations of differentially private algorithms
- **Challenges.** [Ilya Mironov '12...]
 - Keep track of the floating point error
 - Generate the actual privacy loss given the floating point error
- **Methodologies.** [NDA+.'19, BGGHS.'16, YGSBPA.'17...]
 - Design verification system based on the logic of probabilistic coupling and proof techniques in Coq.

Methodologies

prog:
floating point
implementation

input

Programming
logic

output

(ϵ, δ) -dp
or not

- Thanks!